

SingleRAN

BBU Interconnection Feature Parameter Description

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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 01 (2025-03-10).

Technical Changes

Change Description	Parameter Change
Added information about the UMPThb. For details, see 4.1.1.1 UMPT+UMPT Interconnection .	None

Editorial Changes

None

2 About This Document

This document applies to macro base stations including the BTS3900, BTS3900L, BTS3900A, BTS3900AL, BTS3900C, and DBS3900, as well as BTS5900, BTS5900A, BTS5900L, DBS5900, and DBS5900A.

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to GSM, UMTS, LTE FDD, LTE TDD, LTE NB-IoT, and NR. This feature works the same way with each of these RATs.

3 Overview

3.1 Introduction

With the growth of the number of UEs and traffic volume, and the introduction of new services, the capacity of the base station deployed with a single BBU cannot meet the service requirements. You can add BBUs and boards, and enable BBU interconnection for fast capacity expansion of the base station.

BBU infrastructure interconnection enables two BBUs to be connected through a BBU interconnection signal cable, serving the same base station. In infrastructure interconnection, the UMPT or UCIU in one BBU is interconnected with the UMPT in the other BBU by using a BBU interconnection signal cable to transmit control data, clock signals, and transmission data between the two BBUs, thereby implementing inter-BBU coordination between main control boards. Infrastructure interconnection enables multiple modes such as GU+L or G*U*L+G*U*L to be simultaneously deployed in different BBUs in one site. It also allows inter-BBU RAT introduction such as from GU to GU+L.

In the current version, the BBU3900/BBU3910/BBU5900/BBU5910/BBU5900A supports BBU interconnection, but the BBU3910A/BBU3910C/BBU5900C/BookBBU5901/BookBBU5901a does not. If two single-BBU base stations are reconstructed into one site through BBU interconnection, smooth upgrades are not supported.

This document uses the BBU3900/BBU3910/BBU5900/BBU5910 as an example to show BBU network topologies. BBU interconnection topologies of the BBU5900A are the same as those of other BBUs (BBU3900/BBU3910/BBU5900/BBU5910) except for the number of supported slots and board types. Therefore, this document does not describe BBU interconnection topologies of the BBU5900A.

For the convenience of subsequent descriptions, the conventions of BBU networking scenario meanings in this document are as follows:

- GU: In separate-MPT scenarios, two main control boards work in GSM and UMTS modes, respectively. The meanings of UL and GL are similar.
- G*U: In co-MPT scenarios, a main control board works in GSM and UMTS modes. The asterisk (*) indicates that co-MPT is implemented for the modes before and after the asterisk (*). The meaning of G*U*L is similar.

- G[U*L]: In hybrid-MPT scenarios, one main control board works in GSM mode. The other main control board works in UMTS and LTE modes at the same time. The meaning of G[L*N] is similar.
- GU+L: In dual-BBU scenarios, one BBU works in GSM and UMTS modes and the other BBU works in LTE mode. The modes before and after the plus sign (+) are deployed in different BBUs. The meanings of GL+U and GL+N are similar.
- GU+U: In dual-BBU scenarios, one BBU works in GSM and UMTS modes and the other BBU works in UMTS mode, indicating the UMTS inter-BBU deployment. The meanings of GU+UL and G[U*L]+U*L are similar.
- GU->GU+L: indicates the evolution from single BBU to two BBUs. Before the evolution, the BBU works in GSM and UMTS modes. After the evolution, the two BBUs work in GU dual mode and LTE mode, respectively. The sign (->) indicates the evolution from modes before the sign to modes after the sign.

3.2 Related Concepts

Root BBU and Leaf BBU

Based on the topologies between interconnected BBUs, the interconnected BBUs are classified into the root BBU and leaf BBU.

- Root BBU
The BBU without an upper-level BBU is called the root BBU, which is also called the primary BBU or primary subrack. This BBU is referred to as the root BBU in this document.
- Leaf BBU
A BBU other than the root BBU is called the leaf BBU, which is also called the secondary BBU or secondary subrack. This BBU is referred to as the leaf BBU in this document.

When this document describes BBU interconnection scenarios, the root BBU is put before the leaf BBU by default. For example, GU+L indicates that the BBU working in GU mode is the root BBU and the BBU working in LTE mode is the leaf BBU.

Intra-RAT Inter-BBU and Inter-RAT Inter-BBU

Based on the deployment of hardware resources (such as main control boards and BBPs) of different modes in interconnected BBUs, BBU interconnection can be deployed in two methods:

- Inter-RAT inter-BBU
In inter-RAT inter-BBU deployment, two BBUs are deployed with hardware resources of different modes. For example, in N+G*U*L scenarios, one BBU works in NR mode and the other BBU works in GSM, UMTS, and LTE modes.
In inter-RAT inter-BBU scenarios, the management plane supports TLS and non-TLS communications. TLS 1.3 is supported. TLS communication is described as follows:
 - TLS communication depends on the certificate service function. To ensure normal exchange of management plane data between RATs during

- startup, TLS communication is not enabled if the certificate has not yet taken effect.
- If the version of one RAT is earlier than SRAN18.1 or one RAT is carried on a non-UMPT main control board, TLS communication is not enabled.
- Intra-RAT inter-BBU
In intra-RAT inter-BBU deployment, two BBUs are both deployed with single-mode or co-MPT hardware resources. For example, in G[U*L]+U*L scenarios, UL co-MPT hardware resources are deployed in both BBUs, which is the UL inter-BBU deployment.
In intra-RAT inter-BBU scenarios, the management plane supports TLS and non-TLS communications. TLS 1.3 is supported.

Primary Main Control Board and Secondary Main Control Board

In intra-RAT inter-BBU scenarios, the main control board of the mode must be deployed in both BBUs. The main control boards in the root BBU and leaf BBU are called the primary main control board and secondary main control board, respectively. For example, in G[U*L]+U*L scenarios, UMPT_UL boards must be deployed in both BBUs. The UMPT_UL boards in the root BBU and leaf BBU are the primary main control board and secondary main control board, respectively.

When a main control board is operating properly, its ACT indicator shows whether it serves as the primary or secondary main control board. If the ACT indicator is steady on, the board is the primary main control board. If the ACT indicator is steady off, the board is the secondary main control board.

3.3 NE Name for BBU Interconnection

In BBU interconnection scenarios, the NE type of a base station varies with BBU type (BBU39X0/BBU5900/BBU5910/BBU5900A) and interconnection mode (inter-RAT interconnection or intra-RAT interconnection).

Inter-RAT Interconnection

The BBU39X0 and BBU59X0 support inter-RAT interconnection between them.

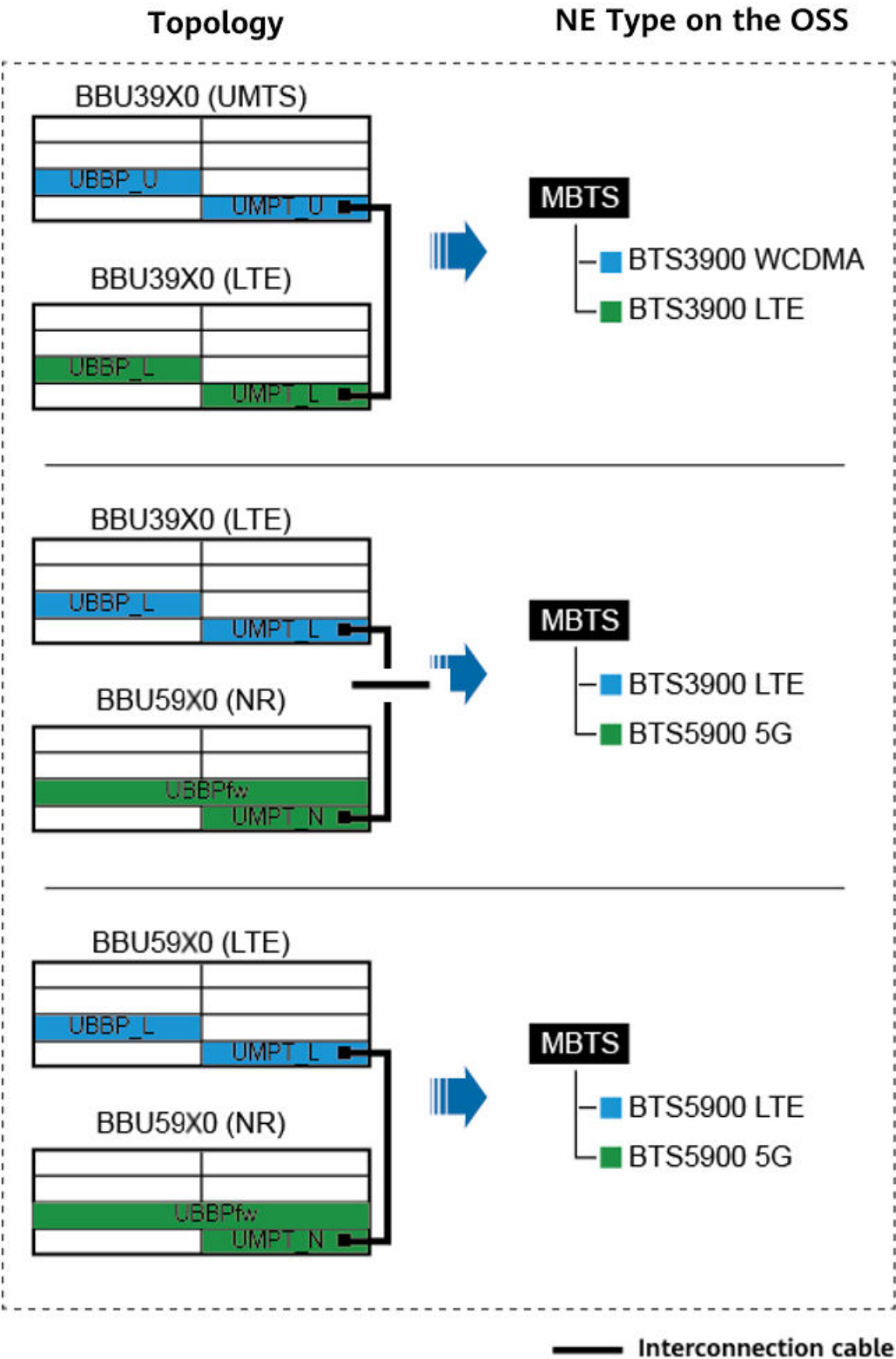
- When a BBU39X0 and another BBU39X0 form a separate-MPT multimode base station through inter-RAT interconnection, the NE types corresponding to the two BBUs are both BTS3900, which can be BTS3900 WCDMA, BTS3900 LTE, BTS3900 NR, or BTS3900.
- When a BBU59X0 and a BBU59X0 form a separate-MPT multimode base station through inter-RAT interconnection, the NE types corresponding to the two BBUs are both BTS5900, which can be BTS5900 WCDMA, BTS5900 LTE, BTS5900 NR, or BTS5900.
- When a BBU59X0 and a BBU39X0 form a separate-MPT multimode base station through interconnection, the NE type corresponding to the BBU59X0 is BTS5900 and the NE type corresponding to the BBU39X0 is BTS3900.

The BBU5900A supports inter-RAT interconnection with the BBU39X0 or BBU59X0, but does not support inter-RAT interconnection with another BBU5900A.

- When a BBU5900A and a BBU59X0 form a separate-MPT multimode base station through inter-RAT interconnection, the NE types corresponding to the two BBUs are both BTS5900.
- When a BBU5900A and a BBU39X0 form a separate-MPT multimode base station through interconnection, the NE type corresponding to the BBU5900A is BTS5900 and the NE type corresponding to the BBU39X0 is BTS3900.

The following figure shows an example.

Figure 3-1 Example for inter-RAT interconnection



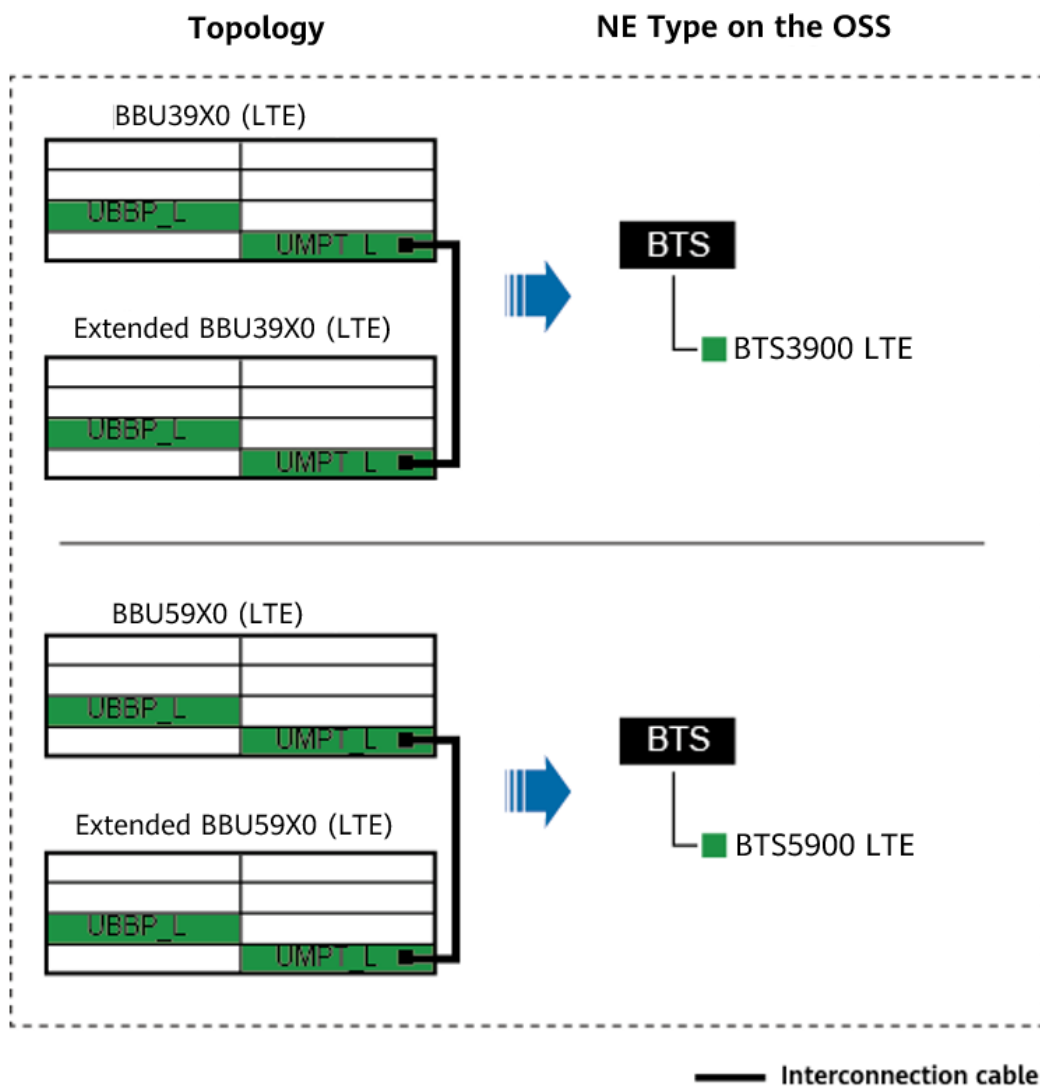
Intra-RAT Interconnection

Intra-RAT interconnection is available between two BBU39X0 modules or two BBU59X0 modules. The BBU5900A does not support intra-RAT interconnection.

- When two BBU39X0s form a base station through intra-RAT interconnection, the NE types of the two BBUs are the same, which can be BTS3900 WCDMA, BTS3900 LTE, or BTS3900.
- When a BBU59X0 and a BBU59X0 form a base station through intra-RAT interconnection, the NE types of the two BBUs are the same, which can be BTS5900 WCDMA, BTS5900 LTE, or BTS5900.

The following figure shows an example.

Figure 3-2 Example for intra-RAT interconnection



4 Infrastructure Interconnection

4.1 Principles

4.1.1 Interconnection Methods

Based on different interconnection boards, two BBUs can be interconnected using the UMPT+UMPT or UCIU+UMPT method. For ease of description, the former is referred to as UMPT+UMPT interconnection, and the latter is referred to as UCIU+UMPT interconnection.

UMPT+UMPT interconnection is recommended.

4.1.1.1 UMPT+UMPT Interconnection

This section describes the cable connections, data configurations, and application restrictions of UMPT+UMPT interconnection.

NOTE

BBU5900As cannot be interconnected.

Combinations Supported by the Root BBU and Leaf BBU

In UMPT+UMPT interconnection, the root BBU and leaf BBU support the following combinations:

NOTE

The UMPT_{Ha}/UMPT_{Hb} cannot work with the BBU39X0.

- Scenario 1: No BBU5910 is configured. The root BBU or leaf BBU can be a BBU3900, BBU3910, BBU5900, or BBU5900A.

Infrastructure Interconnection Method	Root BBU	Leaf BBU
UMPT+UMPT	BBU3900	BBU3900

Infrastructure Interconnection Method	Root BBU	Leaf BBU
UMPT+UMPT	BBU3900	BBU3910
UMPT+UMPT	BBU3900	BBU5900
UMPT+UMPT	BBU3900	BBU5900A
UMPT+UMPT	BBU3910	BBU3900
UMPT+UMPT	BBU3910	BBU3910
UMPT+UMPT	BBU3910	BBU5900
UMPT+UMPT	BBU3910	BBU5900A
UMPT+UMPT	BBU5900	BBU3900
UMPT+UMPT	BBU5900	BBU3910
UMPT+UMPT	BBU5900	BBU5900
UMPT+UMPT	BBU5900	BBU5900A
UMPT+UMPT	BBU5900A	BBU3900
UMPT+UMPT	BBU5900A	BBU3910
UMPT+UMPT	BBU5900A	BBU5900

- Scenario 2: A BBU5910 is configured and interconnected with another BBU5910, BBU5900, or BBU5900A. In this scenario, the root BBU or leaf BBU can be a BBU5910, BBU5900, or BBU5900A.

Infrastructure Interconnection Method	Root BBU	Leaf BBU
UMPT+UMPT	BBU5900	BBU5910
UMPT+UMPT	BBU5900A	BBU5910
UMPT+UMPT	BBU5910	BBU5900
UMPT+UMPT	BBU5910	BBU5900A
UMPT+UMPT	BBU5910	BBU5910

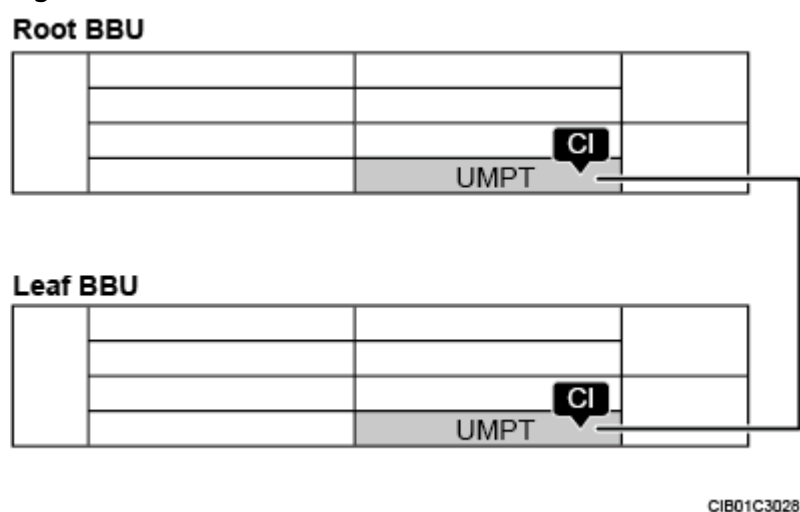
- Scenario 3: A BBU5910 is configured and interconnected with a BBU3910 or BBU3900. In this scenario, the BBU5910 must be the root BBU.

Infrastructure Interconnection Method	Root BBU	Leaf BBU
UMPT+UMPT	BBU5910	BBU3900
UMPT+UMPT	BBU5910	BBU3910

Cable Connections

Figure 4-1 shows the cable connections in UMPT+UMPT interconnection.

Figure 4-1 Cable connections in UMPT+UMPT interconnection



The interconnection port is the CI port on the UMPT. The interconnection signal cables are classified into electrical cables and optical fibers. The cabling requirements are as follows:

- Using an SFP electrical cable
The cable end with a Master color-coding label is connected to the CI port on the UMPT in the root BBU, and the other end is connected to that in the leaf BBU.
- Using single-fiber bidirectional optical modules and a single-mode optical fiber
Single-fiber bidirectional optical modules must be used in pairs. The optical module (blue puller) with a TX wavelength of 1270 nm and an RX wavelength of 1330 nm is installed in the CI port on the UMPT in the root BBU. The optical module (purple puller) with a TX wavelength of 1330 nm and an RX wavelength of 1270 nm is installed in the CI port on the UMPT in the leaf BBU.

In UMPT+UMPT interconnection scenarios, 10G optical modules are required for the CI ports. (The item code is subject to the configurator.) If different types of optical modules are used together or optical modules whose rate exceeds the specifications are used, performance risks such as alarms, bit errors, and link disconnection may occur.

Data Configurations

After the BBU interconnection signal cable is connected, you need to configure inter-BBU interconnection links on each single mode side and the co-MPT side of the base station, and specify the positions of the root BBU, leaf BBU, and interconnection ports. On the GSM (eGBTS), UMTS, LTE, NR, and co-MPT sides, the **CTRLLNK** MO is used to configure an inter-BBU interconnection link. On the GSM (GBTS) side, the **BTSCTRLNK** MO is used to configure an inter-BBU interconnection link. For example, in GU+L scenarios, the UMPT_U and UMPT_L are connected. You need to configure the **BTSCTRLNK** MO on the GSM (GBTS) side, and the **CTRLLNK** MO on both the UMTS and LTE sides. In G*U*L+N scenarios, the UMPT_GUL and UMPT_N are connected. You need to configure the **CTRLLNK** MO on both the G*U*L co-MPT side and NR side. In U+U scenarios, the UMPT_U and UMPT_U are connected. You only need to configure the **CTRLLNK** MO on the UMTS side of the root BBU.

Only one interconnection signal cable can be connected between two UMPT boards. Only one inter-BBU interconnection link is configured on each single mode side or the co-MPT side.

NOTE

- If the root BBU and leaf BBU are installed in the same cabinet, the cabinet numbers (generally 0) and cabinet types must be the same for the root BBU and leaf BBU during BBU data configuration. Otherwise, an alarm is reported.
- If the root BBU and leaf BBU are installed in different cabinets, the cabinet numbers (0 recommended for the root BBU) must be different for the root BBU and leaf BBU. In this case, the cabinet types can be different for them.

Application Restrictions

Different types of UMPT boards (UMPTb/UMPTe/UMPTg/UMPTga/UMPTHa/UMPTb) can be interconnected. [Table 4-1](#) lists the application restrictions on UMPT+UMPT interconnection.

Table 4-1 Application restrictions on UMPT+UMPT interconnection

Category	Application Restrictions
BBUs and boards	• When the clock mutual lock optimization (specified by the BTSOTHPARA.ClkMutualLockOptSw parameter) is enabled on the base station controller side, the GTMUc is a separate-MPT board in the subrack but not the interconnection board. The GTMUc can be configured in the leaf BBU. If NR is enabled, the GTMUc is preferentially deployed in the leaf BBU. • If the leaf BBU is configured with a GTMUc, the leaf BBU supports RRU cascading. • If the NR mode is deployed, it is recommended that the BBU deployed with NR be the root BBU. • In UMPT+UMPT interconnection mode, neither the root BBU nor the leaf BBU supports main control board backup. The BBU interconnection signal cable cannot be connected to the standby main control board. • Intra-RAT interconnection is not supported in UMPTha/UMPTb+UMPTb interconnection. • In GSM intra-RAT inter-BBU scenarios, GSM intra-band RRUs cannot be deployed in different BBUs.
Transmission	The root BBU must provide a transmission port in intra-RAT inter-BBU deployment.

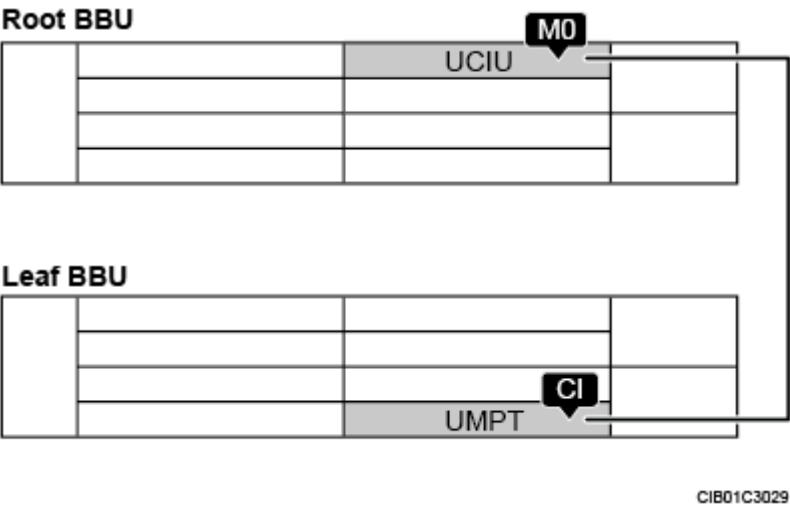
4.1.1.2 UCIU+UMPT Interconnection

In UCIU+UMPT interconnection, the root BBU must be a BBU3900, and the leaf BBU can be a BBU3900/BBU3910/BBU5900/BBU5900A. This section describes the cable connections, data configurations, and application restrictions of UCIU+UMPT interconnection.

Cable Connections

[Figure 4-2](#) shows the cable connections in UCIU+UMPT interconnection.

Figure 4-2 Cable connections in UCIU+UMPT interconnection



The cabling requirements are as follows:

- The BBU interconnection signal cable must be a multimode optical fiber. In addition, multimode optical modules are used.
- The interconnection port can be any port from M0 to M4 on the UCIU or the CI port on the UMPT.
- If only one UMPT is configured in the leaf BBU, you are advised to connect the CI port on this UMPT to the M0 port on the UCIU in the root BBU. If two UMPT boards (active/standby) are configured in the leaf BBU, you are advised to connect the CI ports on these UMPT boards to the M0 and M1 ports on the UCIU in the root BBU, respectively.

Data Configurations

After the BBU interconnection signal cable is connected, you need to configure inter-BBU interconnection links (also called infrastructure interconnection links) on each single mode side and the co-MPT side of the base station, and specify the positions of the root BBU, leaf BBU, and interconnection ports. On the GSM (eGBTS), UMTS, LTE,, NR, and co-MPT sides, the **CTRLLNK** MO is used to configure an inter-BBU interconnection link. On the GSM (GBTS) side, the **BTSCTRLLNK** MO is used to configure an inter-BBU interconnection link. For example, in GU+L scenarios, you need to configure the **BTSCTRLLNK** MO on the GSM (GBTS) side, and the **CTRLLNK** MO on both the UMTS and LTE sides.

If the leaf BBU is configured with two UMPT boards, the two UMPT boards are interconnected with the UCIU board, and you need to configure two inter-BBU interconnection links on each single mode side or the co-MPT side. Otherwise, you need to configure one inter-BBU interconnection link on each single mode side or the co-MPT side.

 NOTE

- If the root BBU and leaf BBU are installed in the same cabinet, the cabinet numbers (generally 0) and cabinet types must be the same for the root BBU and leaf BBU during BBU data configuration. Otherwise, an alarm is reported.
- If the root BBU and leaf BBU are installed in different cabinets, the cabinet numbers (0 recommended for the root BBU) must be different for the root BBU and leaf BBU. In this case, the cabinet types can be different for them.

Application Restrictions

Table 4-2 lists the restrictions on UCIU+UMPT interconnection.

Table 4-2 Application restrictions on UCIU+UMPT interconnection

Category	Application Restrictions
BBU subracks and boards	• The UCIU can be installed only in the BBU3900, and the BBU must be the root BBU. • If GSM is to be deployed, all GSM boards (such as main control boards, BBPs, and interface boards) must be configured in the root BBU.
Transmission	The root BBU must provide a transmission port in intra-RAT inter-BBU deployment.

4.1.2 Application Scenarios

Table 4-3 and **Table 4-4** list the application scenarios supported by infrastructure interconnection in the current version.

Table 4-3 Application scenarios of infrastructure interconnection (inter-RAT inter-BBU)

Application Scenario (Root BBU RAT+Leaf BBU RAT)	Infrastructure Interconnection Method
GU+L	• UCIU+UMPT • UMPT+UMPT
GL+U	• UCIU+UMPT • UMPT+UMPT
L*N+GU	UMPT+UMPT
L*N+G*U	UMPT+UMPT
L*N+U	UMPT+UMPT
LN+GU	UMPT+UMPT
LN+U	UMPT+UMPT

Application Scenario (Root BBU RAT+Leaf BBU RAT)	Infrastructure Interconnection Method
N+G*U*L	UMPT+UMPT
N+L	UMPT+UMPT
N+GL	UMPT+UMPT
N+UL	UMPT+UMPT
L+N	UMPT+UMPT
GL+N	UMPT+UMPT
UL+N	UMPT+UMPT
U+LN	UMPT+UMPT
G*U*L+N	UMPT+UMPT

Table 4-4 Application scenarios of infrastructure interconnection (intra-RAT inter-BBU)

Application Scenario (Root BBU RAT+Leaf BBU RAT)	Infrastructure Interconnection Method
GU+UL	UCIU+UMPT
GU+U	• UCIU+UMPT • UMPT+UMPT
U+U	• UCIU+UMPT • UMPT+UMPT
G[U*L]+U*L	• UCIU+UMPT • UMPT+UMPT
G*U*L+G*U*L	• UCIU+UMPT • UMPT+UMPT

 NOTE

If two co-MPT base stations are interconnected, the **WMEXTENSION** parameter must be set to different values for the two base stations.

For example, in L*N+G*U scenarios, it is recommended that the parameter be set to **COMM2** for L*N and **COMM** for G*U. If the parameter is set to the same value, alarms will be generated on the base station and services will be abnormal.

4.1.3 Supported Features and Functions

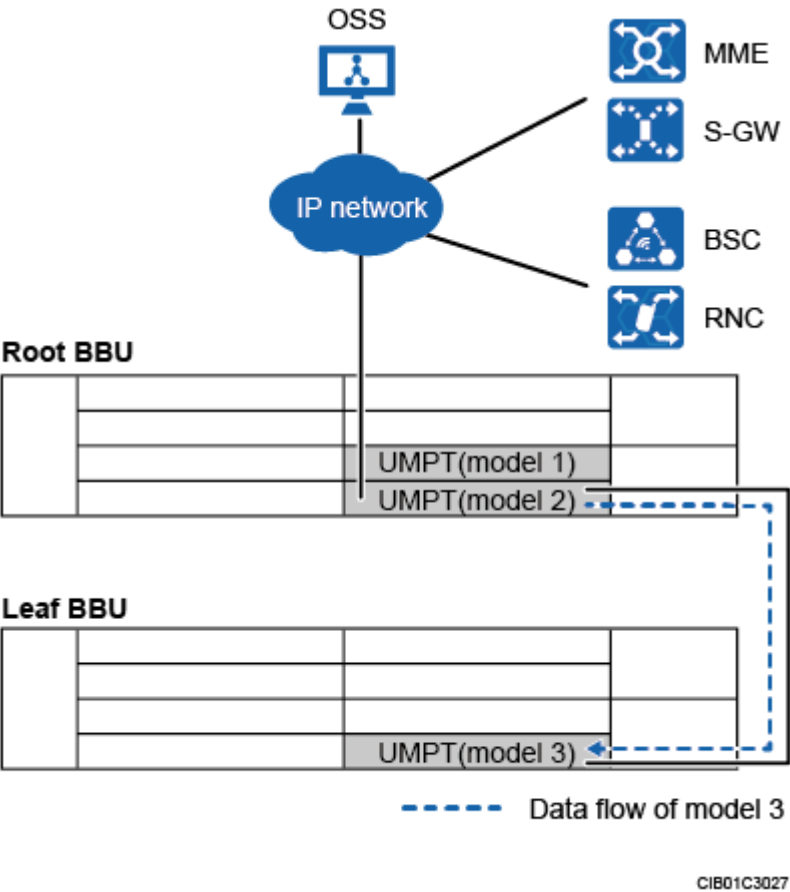
After infrastructure interconnection is implemented between two BBUs, the BBUs can exchange control data, clock signals, and transmission data. Therefore, the

BBUs can support inter-BBU common transmission (co-transmission), common clock, and software-defined radio (SDR).

Inter-BBU Co-Transmission

With inter-BBU co-transmission, one BBU provides a co-transmission port to connect to the transmission network and provides a transmission channel for the other BBU. Two interconnected BBUs support co-transmission between any RATs. Generally, the co-transmission port should be provided by the transmission board with the highest specifications in the root BBU or leaf BBU. For example, in the L*N+G*U interconnection scenario, the UMPT (LN) must provide the co-transmission port. [Figure 4-3](#) shows an example of inter-BBU co-transmission. For details, see *Common Transmission Feature Parameter Description* in the feature documentation of the corresponding RAT, for example, *Common Transmission Feature Parameter Description* in 5G RAN feature documentation.

Figure 4-3 Inter-BBU co-transmission example



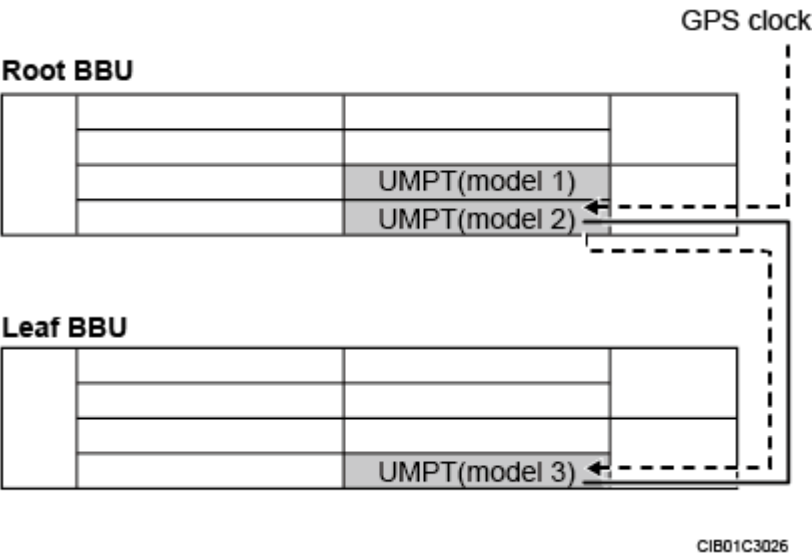
NOTE

When inter-BBU co-transmission is enabled, the total number of LTE FDD and LTE TDD BBPs configured in the two BBUs cannot exceed 10 if the UMPTb connects to the transmission network.

Inter-BBU Common Clock

With inter-BBU common clock, one BBU obtains the external shared clock source and provides the clock source for the other BBU. Two interconnected BBUs support common clock between any RATs. The external clock source can be obtained by the root BBU or leaf BBU. **Figure 4-4** shows an example of inter-BBU common clock. For details, see *Common Clock Feature Parameter Description* in the feature documentation of the corresponding RAT, for example, *Common Clock Feature Parameter Description* in 5G RAN feature documentation.

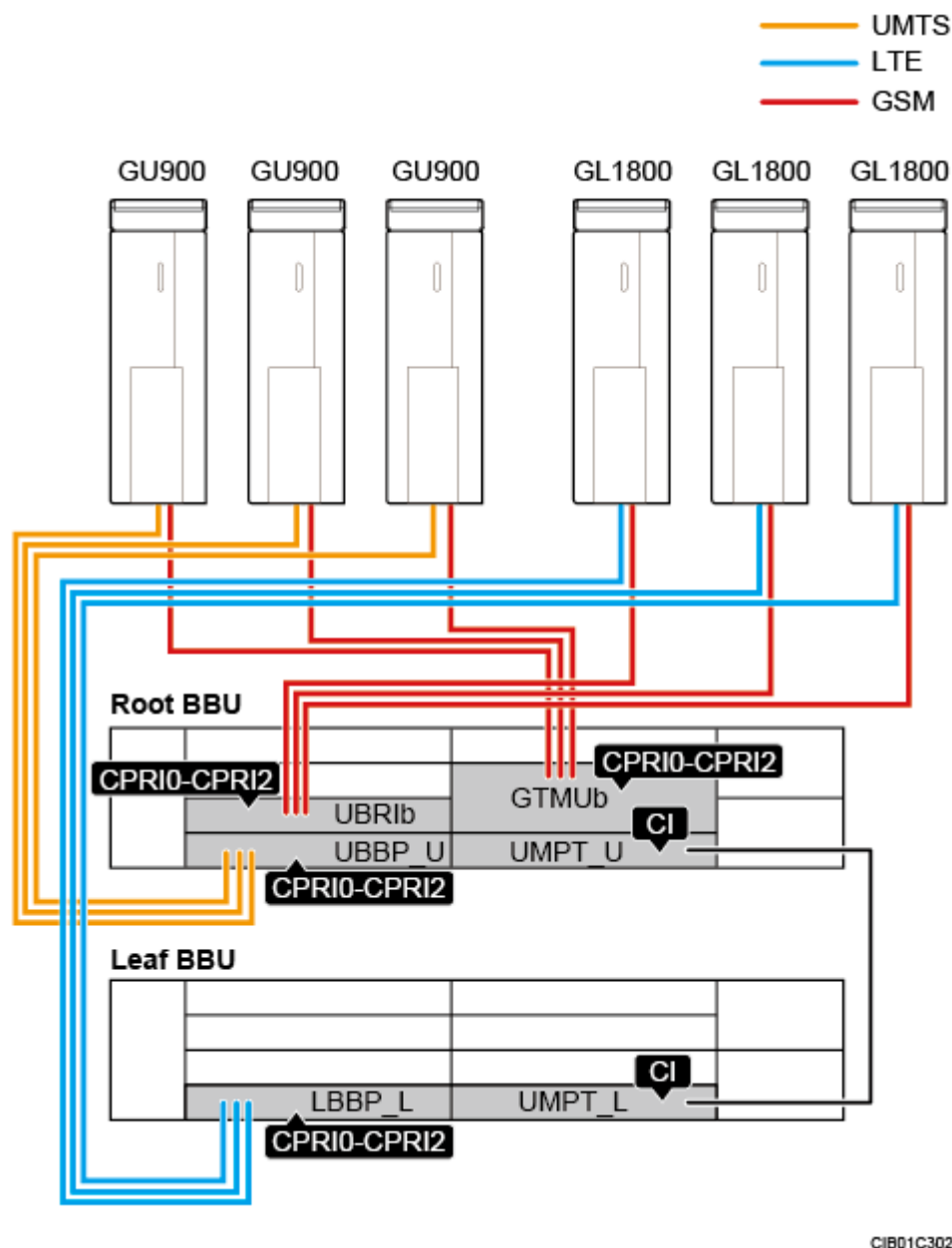
Figure 4-4 Inter-BBU common clock example



Inter-BBU SDR

In inter-BBU SDR, a multimode RF module is connected to two interconnected BBUs to implement flexible deployment of a multimode base station. **Figure 4-5** shows an example of inter-BBU SDR. For details, see *RF Unit and Topology Management Feature Parameter Description* in the feature documentation of the corresponding RAT, for example, *RF Unit and Topology Management Feature Parameter Description* in 5G RAN feature documentation.

Figure 4-5 Inter-BBU SDR



As shown in the preceding figure, RF modules supporting GL are connected to both the root BBU and leaf BBU.

Inter-BBU SDR requires that the clock synchronization mode be configured according to the following rules:

- If the root BBU uses frequency synchronization, the leaf BBU must use frequency synchronization.
- If the root BBU uses time synchronization, the leaf BBU can use either time or frequency synchronization.

4.2 Network Analysis

4.2.1 Benefits

Infrastructure interconnection enables multiple modes to share a base station, saving site deployment resources.

4.2.2 Impacts

The impacts between this function and other functions are described in [Function Impacts](#).

This function has no impact on the network.

4.3 Requirements

4.3.1 Licenses

None

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

UMPT+UMPT Interconnection

When UMPT+UMPT interconnection is used, there is no special requirement for software versions.

UCIU+UMPT Interconnection

When UCIU+UMPT interconnection is used, the software version requirements are as follows:

- SRAN7.0 V100R007C00SPC280 or later if UMPTb boards are used for BBU interconnection.
- SRAN11.1 V100R011C10SPC150 or later if UMPTe boards are used for BBU interconnection.
- SRAN15.1 V100R015C10SPC150 or later if UMPTg boards are used for BBU interconnection.

In UCIU+UMPT interconnection, the UCIU is controlled by only one RAT. Otherwise, ALM-26274 Inter-System Board Object Configuration Conflict is

reported. You are advised to comply with the following rules to set the RAT controlling the UCIU:

- For inter-BBU deployment of a single RAT, this RAT is preferred for controlling the UCIU.
- If a RAT is added to an existing base station, you are not advised to change the RAT controlling the UCIU. However, you can modify the RAT controlling the UCIU according to scenario-specific recommendations.
- In other scenarios, deploy the RAT controlling a UCIU following a descending order of priority: GSM > UMTS > LTE > NR.

 NOTE

During base station commissioning, it is recommended that the RAT controlling the UCIU be deployed first. This will allow UCIU-related alarms to be reported in time. However, the RAT deployment order remains unchanged regardless of the deployment status of the RAT controlling the UCIU.

4.3.2.1 Prerequisite Functions

None

4.3.2.2 Mutually Exclusive Functions

None

4.3.2.3 Function Impacts

None

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

UMPT+UMPT Interconnection

For UMPT+UMPT interconnection, select an interconnection signal cable based on the distance between the root BBU and leaf BBU.

- If the distance is less than or equal to 5 m, use the Huawei SFP electrical cable.
- If the distance is greater than 5 m, use Huawei single-fiber bidirectional optical modules and single-mode optical fibers. By default, the length of the delivered single-mode optical fiber is 10 m. The maximum length of a single-mode optical fiber is 100 m.

For details about the BBU interconnection signal cable and its installation method, refer to the section on cable installation or BBU interconnection signal cable installation in the related installation guide, located at the following path: **3900 & 5900 Series Base Station Product Documentation (Hardware Installation) > Base Station Installation Guide > DBS3900 & DBS5900 Installation Guide**.

UCIU+UMPT Interconnection

The maximum length of a multimode optical fiber is 10 m. If the distance between two BBUs exceeds 10 m, contact Huawei technical support.

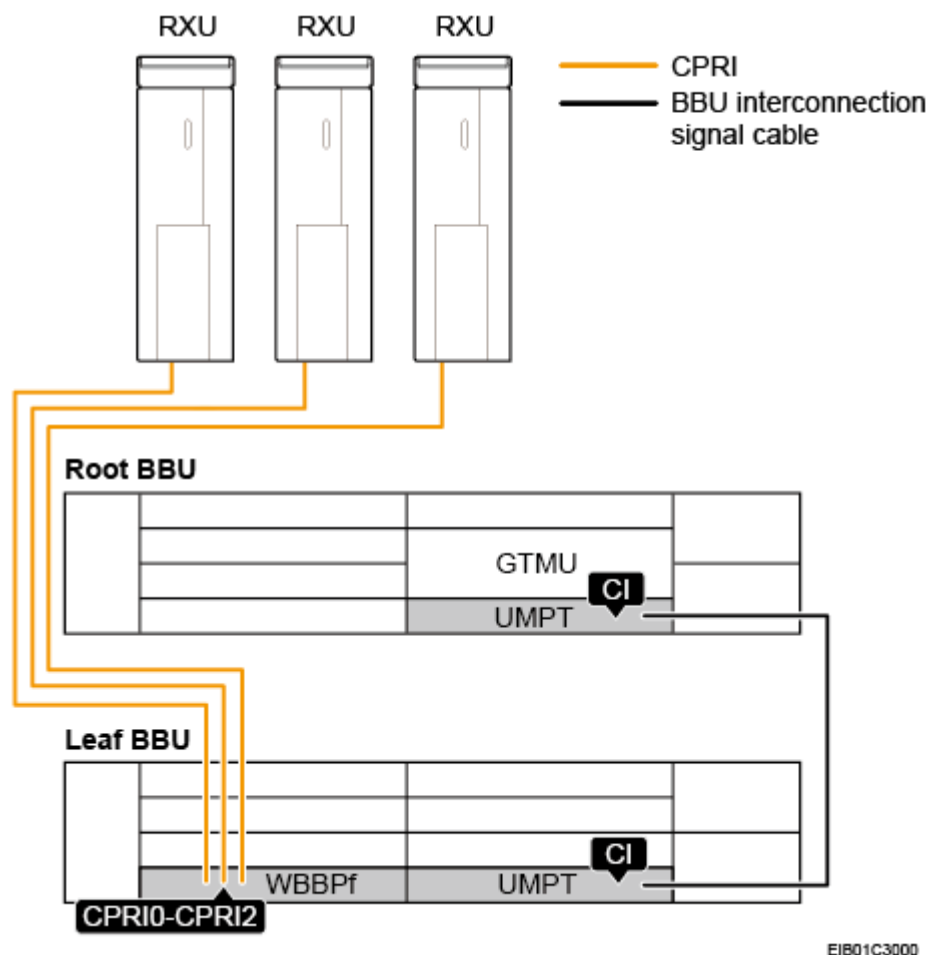
4.3.4 Others

With infrastructure interconnection enabled, a base station has the following requirements for CPRI-based topologies, transmission networking, and clocks.

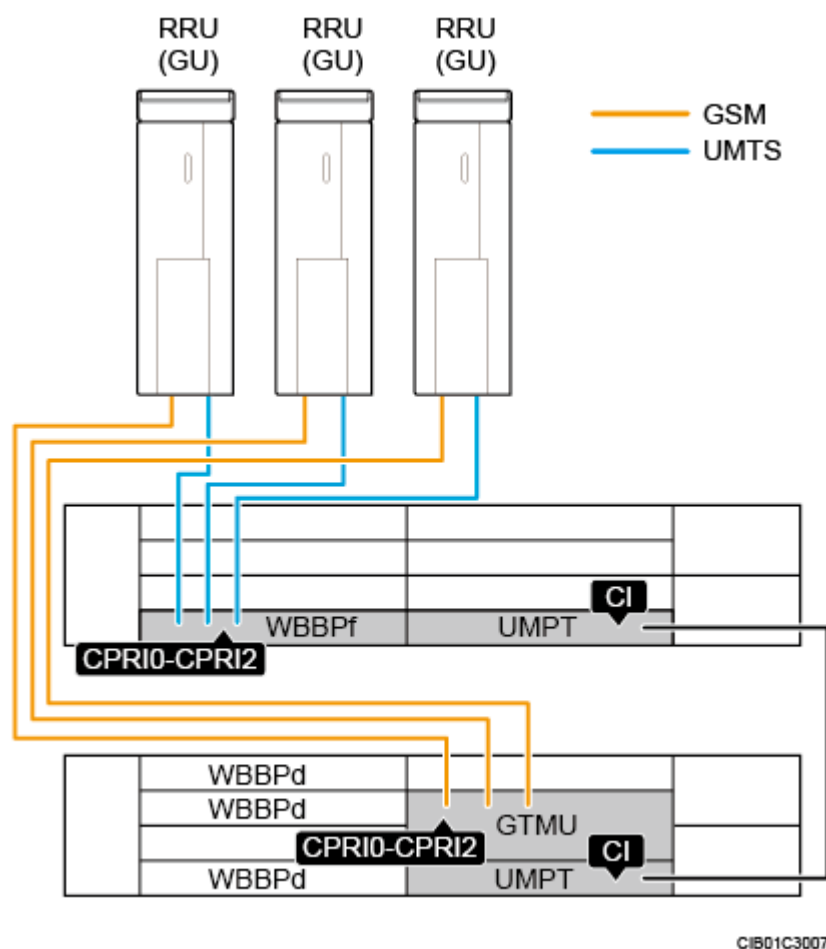
CPRI-based Topologies

In UMPT+UMPT interconnection, only the star topology, not the chain or ring topology, is supported for the leaf BBU when the root BBU is configured with a GTMU. [Figure 4-6](#) shows the CPRI-based topology when single-mode RF modules are used.

Figure 4-6 CPRI-based topology requirements (single-mode RF modules are used)



If multimode RF modules are used, the dual-star CPRI-based topology can be used. [Figure 4-7](#) shows the CPRI-based topology using GU dual-mode RF modules as an example.

Figure 4-7 Dual-star CPRI-based topology (using GU dual-mode RF modules as an example)**NOTE**

For details about the CPRI-based topologies and specifications of CPRI ports, see *RF Unit and Topology Management Feature Parameter Description* and *CPRI MUX Feature Parameter Description* in the feature documentation of the corresponding RAT, for example, *RF Unit and Topology Management Feature Parameter Description* and *CPRI MUX Feature Parameter Description* in 5G RAN feature documentation.

Transmission Networking

Transmission networking must meet the requirements listed in [Table 4-2](#) and [Table 4-1](#). Based on this, multimode base stations use the following two transmission modes when the corresponding requirements are met:

- Independent transmission
When each RAT adopts independent transmission, the transmission networking requirements for each RAT are the same as those for the corresponding single-mode base station. For details about the transmission networking requirements for single-mode base stations, see the following documents:
 - *IP BSS Engineering Guide Feature Parameter Description* for GSM in GBSS feature documentation

- *IP RAN Engineering Guide Feature Parameter Description* for UMTS in RAN feature documentation
- *IP eRAN Engineering Guide Feature Parameter Description* for LTE in eRAN feature documentation
- *IP NR Engineering Guide Feature Parameter Description* for NR in 5G RAN feature documentation
- Co-transmission
In inter-RAT co-transmission, the root BBU or leaf BBU provides a transmission port. For details about the requirements for inter-BBU co-transmission networking, see *Common Transmission Feature Parameter Description* in the feature documentation of the corresponding RAT, for example, *Common Transmission Feature Parameter Description* in 5G RAN feature documentation.

Clocks

After BBUs are interconnected, a multimode base station can obtain clock signals in either of the following ways:

- Independent clock
When each RAT adopts independent clock, the clock requirements for each RAT are the same as those for the corresponding single-mode base station. For details about the clock requirements for single-mode base stations, see *Synchronization Feature Parameter Description* in the feature documentation for GSM, UMTS, LTE, or NR.
- Common clock
When the clock source is shared between RATs, the clock source can be deployed in the root BBU or leaf BBU. For details about the requirements for inter-BBU common clock, see *Common Clock Feature Parameter Description* in the feature documentation of the corresponding RAT, for example, *Common Clock Feature Parameter Description* in 5G RAN feature documentation.

4.4 Operation and Maintenance

4.4.1 When to Use

If the hardware resources of a single BBU cannot meet the service deployment requirements, you are advised to enable infrastructure interconnection.

4.4.2 Required Information

None

4.4.3 Data Configuration

4.4.3.1 Data Preparation

The **CTRLLNK** and **BTSCTRLLNK** MOs are used to configure inter-BBU interconnection links. The key parameters are described in the following table.

Table 4-5 Key parameters related to **CTRLLNK** or **BTSCTRLLNK**

Parameter Name	Parameter ID	Setting Notes
Local Link No.	<i>LN (LTE eNodeB, 5G gNodeB)</i>	Number of an inter-BBU interconnection link
Local Cabinet No.	<i>CN (LTE eNodeB, 5G gNodeB)</i>	Cabinet number of the leaf BBU
Local Subrack No.	<i>SRN (LTE eNodeB, 5G gNodeB)</i>	Subrack number of the leaf BBU
Local Slot No.	<i>SN (LTE eNodeB, 5G gNodeB)</i>	Slot number of the UMPT to which the BBU interconnection signal cable is connected in the leaf BBU. Set this parameter to 6 or 7 based on site requirements.
Upper Cabinet No.	<i>UPCN (LTE eNodeB, 5G gNodeB)</i>	Cabinet number of the root BBU
Upper Subrack No.	<i>UPSRN (LTE eNodeB, 5G gNodeB)</i>	Subrack number of the root BBU
Upper Slot No.	<i>UPSN (LTE eNodeB, 5G gNodeB)</i>	Slot number of the UCIU/UMPT to which the BBU interconnection signal cable is connected in the root BBU. Set this parameter based on site requirements. • The UCIU can be configured in slot 4 or 0. • The UMPT can be configured in slot 6 or 7.
Upper Port No.	<i>UPPT (LTE eNodeB, 5G gNodeB)</i>	Interconnection port number of the UCIU/UMPT to which the BBU interconnection signal cable is connected in the root BBU. Set this parameter based on actual connections. • For the UCIU, the port number ranges from 0 to 4, corresponding to the M0 to M4 ports on the UCIU. • For the UMPT, the port number is 8, corresponding to the CI port on the UMPT.

4.4.3.2 Initial Configuration

Before using MML commands, refer to [4.2.2 Impacts](#) and [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

The MAE-Deployment is recommended for initial configuration of a new base station. The following table lists references for the initial configuration procedures in BBU interconnection based on separate- and co-MPT base stations.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Scenario	Procedure
BBU interconnection for a separate-MPT base station (for example, GL+U and GU+UL)	See Creating a Single Separate-MPT Base Station > Configuring Device Data About the Separate-MPT Base Station > Configuring Cabinet, Subrack, and BBU Data and Configuring BBU Interconnection Links in <i>Separate-MPT Base Station Initial Configuration Guide (MAE-Deployment-based)</i> .
BBU interconnection for a co-MPT base station (for example, G*U*L+G*U*L)	See Creating a Single Co-MPT Base Station > Configuring Device Data About the Co-MPT Base Station > Configuring Cabinet, Subrack, and BBU Data and Configuring BBU Interconnection Links in <i>Co-MPT Base Station Initial Configuration Guide (MAE-Deployment-based)</i> .

4.4.4 Activation Verification

After the initial configuration is complete, you can observe the status of the inter-BBU interconnection link at either the local or remote end.

Local Observation

At the local end, check the status of indicators on the interconnection board to determine whether the inter-BBU interconnection link is operating properly.

- If the Mx (M0 to M4) indicator on the UCIU is green, the inter-BBU interconnection link is operating properly.
- If the CI indicator on the UMPT is green, the inter-BBU interconnection link is operating properly.

NOTE

In UMPT+UMPT interconnection, the interconnection indicator indicates that the interconnection is abnormal if the UMPT in the same BBU as GTMU is not connected to the Master end on the SFP cable.

Remote Observation

NOTE

The status of the inter-BBU interconnection links is updated once an hour. Therefore, if you run the **DSP CTRLINKSTAT** and **DSP BTSCTRLINKSTAT** commands shortly after these links are disconnected and then restored, the command outputs may still show that the link status is unavailable.

Check whether the inter-BBU interconnection link is operating properly.

- Step 1** On the device panel, you can query the master/slave indication of the BBU interconnection signal cable to check whether the cable is correctly connected. The end with a Master color-coding label is yellow.
- Step 2** You can run the **DSP CTRLINKSTAT** command on the GSM (eGBTS), UMTS, LTE, NR, or co-MPT side to check whether the inter-BBU interconnection link is operating properly.
- Step 3** You can run the **DSP BTSCTRLINKSTAT** command on the controller managing the GBTS to check whether the inter-BBU interconnection link is operating properly.

----End

4.4.5 Adjustment

The section provides adjustment examples in the following scenarios where the BBU3900 is used.

NOTE

- Interconnection mode adjustment: Inter-BBU services are interrupted when the interconnection mode is adjusted. For example, the UCIU is replaced by the UMPT for interconnection, the relationship between the primary and secondary BBUs is changed, or the interconnection port is changed. In this case, you need to reset the entire base station to initialize the base station.
- Interconnection state adjustment: For example, when the inter-BBU interconnection in a separate-MPT base station is changed to non-interconnection or from non-interconnection to inter-BBU interconnection, the main control board in the secondary BBU must be reset to ensure that the service functions of the primary and secondary BBUs are normal.

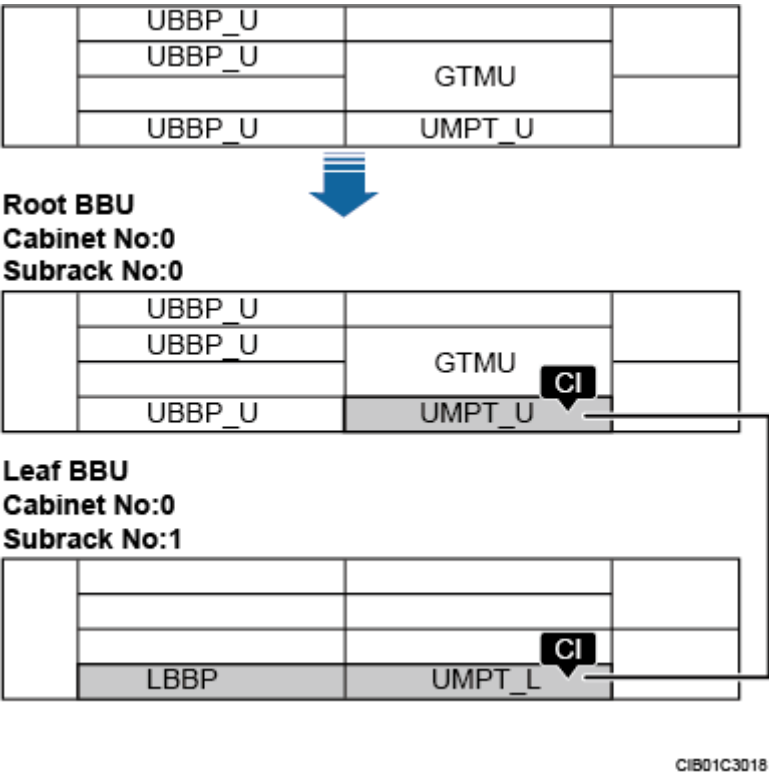
4.4.5.1 GU to GU+L

This section uses adjustment from a GU base station to a GUL base station as an example to describe how to enable infrastructure interconnection.

UMPT+UMPT Interconnection

- The following figure shows the BBU hardware configurations before and after adjustment.

Figure 4-8 Hardware changes during adjustment from a GU base station to a GU+L base station (UMPT+UMPT)



- Preparing new hardware components

Table 4-6 Hardware to be added during adjustment from a GU base station to a GU+L base station (UMPT+UMPT)

New Hardware Component	Quantity
BBU3900 subrack	1
UMPT_L	1
LBBP	1
BBU interconnection signal cable (SFP electrical cable)	1
Other cables, such as the CPRI cable, transmission cable, clock signal cable, power cable, and monitoring signal cable	Depending on the site plan

- Adjusting hardware
 - a. Install a UMPT_L and an LBBP in the leaf BBU. Then, connect the power cables, monitoring signal cables, transmission cables, clock signal cables, and CPRI cables.
 - b. Use a BBU interconnection signal cable to connect the root and leaf BBUs. Connect the cable end with a "Master" color-coding label to the CI

port on the UMPT_U in the root BBU, and the other end to that on the UMPT_L in the leaf BBU.

- Modifying configuration data

You can modify data on the MAE-Deployment. For details, see **Creating a Single Separate-MPT Base Station > Configuring Device Data About the Separate-MPT Base Station > Configuring Cabinet, Subrack, and BBU Data** and **Configuring BBU Interconnection Links** in *Separate-MPT Base Station Initial Configuration Guide (MAE-Deployment-based)*.

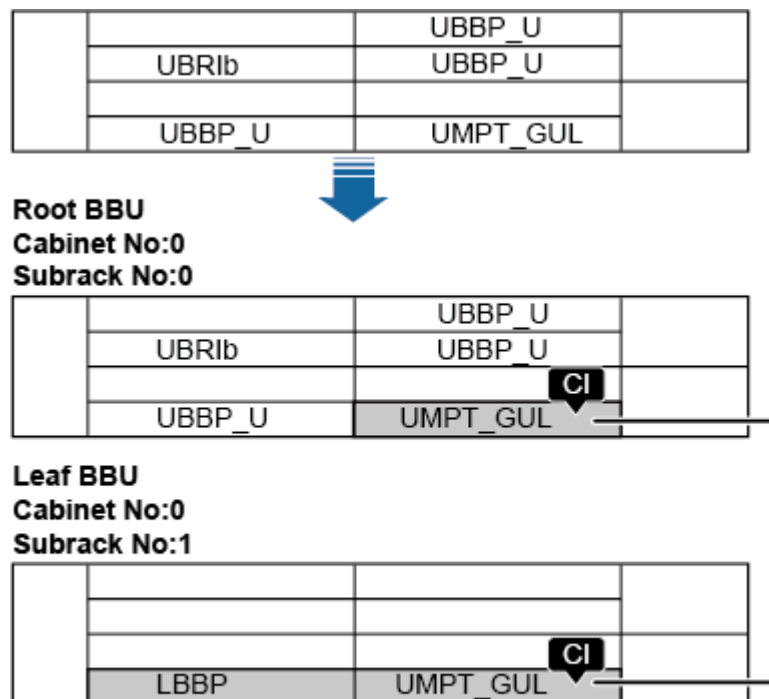
The data reconfiguration procedure using MML commands is as follows:

- a. On the LTE side, run the **ADD CABINET** command to add a cabinet for the leaf BBU.
- b. On the LTE side, run the **ADD SUBRACK** command to add a subrack for the leaf BBU.
- c. On the LTE side, run the **ADD BRD** command to add a UMPT_L for the leaf BBU.
- d. On the LTE side, run the **ADD BRD** command to add an LBBP for the leaf BBU.
- e. On the LTE side, run the **ADD CTRLLNK** command to add an inter-BBU interconnection link.
- f. On the GSM side, run the **ADD BTSCTRLLNK** command to add an inter-BBU interconnection link.
- g. On the UMTS side, run the **ADD CTRLLNK** command to add an inter-BBU interconnection link.

4.4.5.2 G*U*L to G*U*L+G*U*L

This section uses adjustment from a G*U*L base station to a G*U*L+G*U*L base station as an example to describe how to enable infrastructure interconnection.

Figure 4-9 shows the hardware configurations before and after BBUs are interconnected.

Figure 4-9 Hardware changes during adjustment from a G*U*L base station to a G*U*L+G*U*L base station

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- Preparing new hardware components

New Hardware Component	Quantity
BBU3900 subrack	1
UMPT_GUL	1
LBBP	1
BBU interconnection signal cable (SFP electrical cable)	1
Other cables, such as the CPRI cable, transmission cable, clock signal cable, power cable, and monitoring signal cable	Depending on the site plan

- Adjusting hardware
 - Install a UMPT_GUL and an LBBP in the leaf BBU. Then, connect the power cables, monitoring signal cables, transmission cables, clock signal cables, and CPRI cables.
 - Use a BBU interconnection signal cable to connect the root and leaf BBUs. Connect the cable end with a "Master" color-coding label to the CI port on the UMPT_GUL in the root BBU, and the other end to that on the UMPT_GUL in the leaf BBU.

- Modifying configuration data

You can modify data on the MAE-Deployment. For details, see **Creating a Single Co-MPT Base Station > Configuring Device Data About the Co-MPT Base Station > Configuring Cabinet, Subrack, and BBU Data** and **Configuring BBU Interconnection Links** in *Co-MPT Base Station Initial Configuration Guide (MAE-Deployment-based)*.

The data reconfiguration procedure using MML commands is as follows:

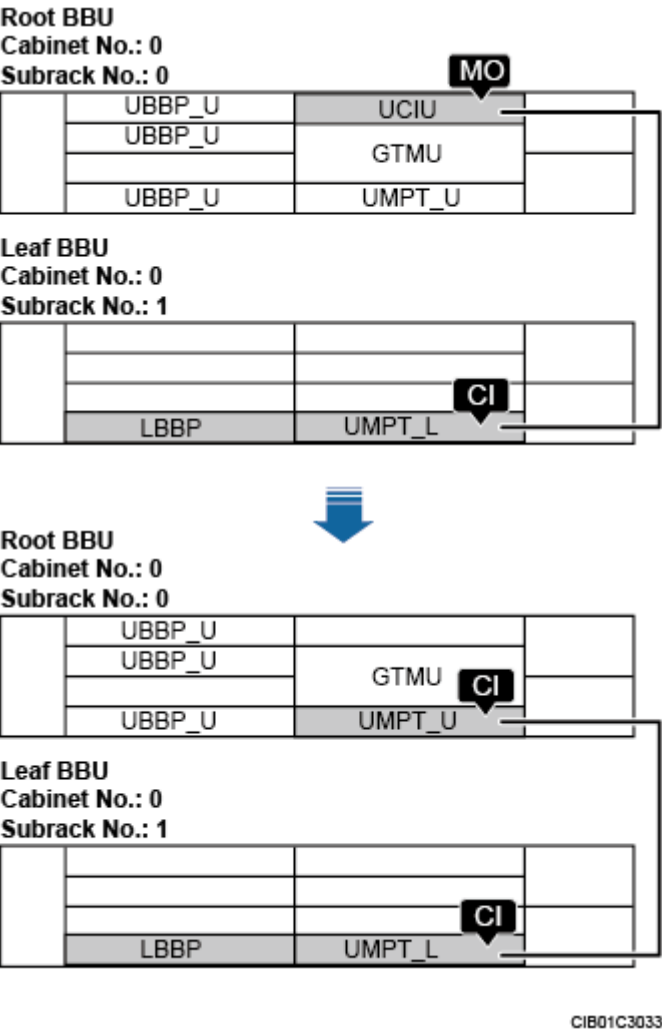
- a. Run the **ADD CTRLINK** command to add an inter-BBU interconnection link.
- b. Run the **ADD SUBRACK** command to add a subrack for the leaf BBU.
- c. Run the **ADD BRD** command to add a UMPT_GUL for the leaf BBU.
- d. Run the **ADD BRD** command to add an LBBP for the leaf BBU.

4.4.5.3 GU+L (UCIU+UMPT to UMPT+UMPT)

This section uses the adjustment from UCIU+UMPT interconnection to UMPT+UMPT interconnection in a GU+L base station as an example.

Figure 4-10 shows the hardware configurations before and after BBUs are interconnected.

Figure 4-10 Hardware changes during adjustment from UCIU+UMPT interconnection to UMPT+UMPT interconnection in a GU+L base station



- Preparing new hardware components

New Hardware Component	Quantity
UMPT	1
BBU interconnection signal cable (SFP electrical cable)	1
- Adjusting hardware
 - Remove the optical fiber between the root and leaf BBUs, optical module from the UCIU in the root BBU, and optical module from the CI port on the UMPT in the leaf BBU.
 - Remove the UCIU from the root BBU.
 - Remove the UMPT_U from the root BBU.
 - Install a new UMPT in the slot that originally housed a UMPT_U. The new UMPT functions as the UMTS main control board.

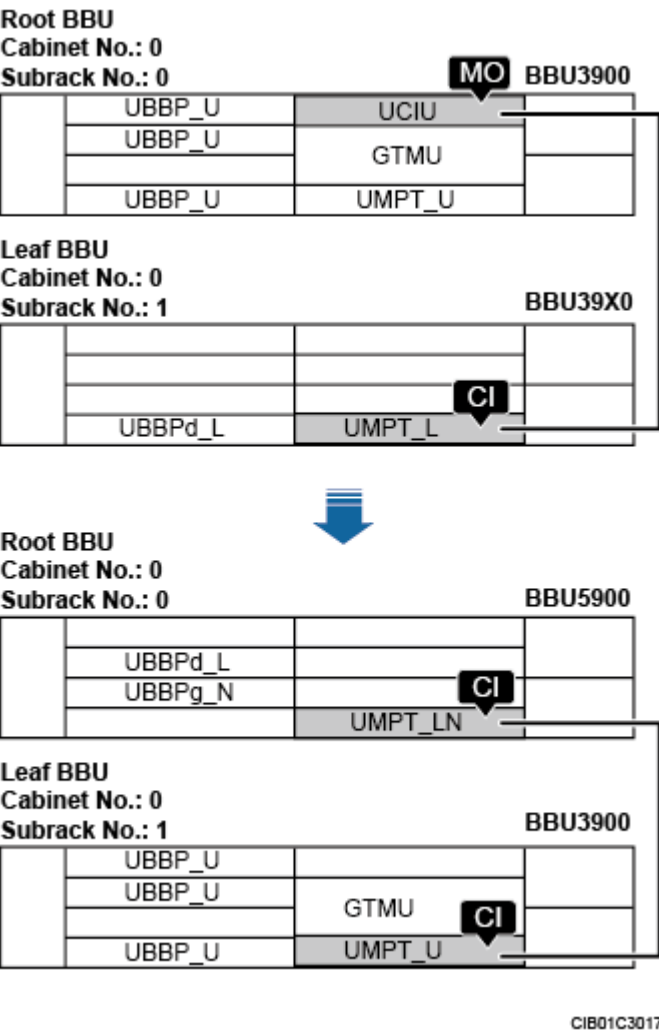
- e. Use a BBU interconnection signal cable to connect the root and leaf BBUs. Connect the cable end with a "Master" color-coding label to the CI port on the UMPT in the root BBU, and the other end to that on the UMPT in the leaf BBU.
- Modifying configuration data
The data reconfiguration procedure using MML commands is as follows:
 - a. On the GSM side, run the **RMV BTSCTRLLNK** command to remove the inter-BBU interconnection link.
 - b. On the GSM side, run the **RMV BTSBRD** command to remove the UCIU from the root BBU.
 - c. On the UMTS side, run the **RMV CTRLLNK** command to remove the inter-BBU interconnection link.
 - d. On the LTE side, run the **RMV CTRLLNK** command to remove the inter-BBU interconnection link.
 - e. On the UMTS side, run the **ADD CTRLLNK** command to add an inter-BBU interconnection link.
 - f. On the LTE side, run the **ADD CTRLLNK** command to add an inter-BBU interconnection link.
 - g. On the GSM side, run the **ADD BTSCTRLLNK** command to add an inter-BBU interconnection link.

4.4.5.4 GU+L to L*N+GU

This section uses adjustment from a GU+L base station using UCIU+UMPT interconnection to a L*N+GU base station using UMPT+UMPT interconnection as an example to describe how to enable infrastructure interconnection.

Figure 4-11 shows the hardware configurations before and after BBUs are interconnected. Before the adjustment, the BBU subrack for GU is the root BBU. After the adjustment, the BBU subrack for L*N is the root BBU.

Figure 4-11 Hardware changes during adjustment from UCIU+UMPT interconnection (GU+L) to UMPT+UMPT interconnection (L*N+GU)



- Preparing new hardware components

New Hardware Component	Quantity
BBU59X0	1, including the power supply and fan
UMPTg_LN	1
UBBPg_N	1
BBU interconnection signal cable (SFP electrical cable)	1

- Adjusting hardware
 - a. Remove the optical fiber between the original root and leaf BBUs, optical module from the UCIU in the original root BBU, and optical module from the CI port on the UMPT in the original leaf BBU.

- b. Remove the UCIU from the original root BBU.
- c. Remove the WMPT from the original root BBU.
- d. Remove the UMPT from the original leaf BBU. Install the UMPT as the UMTS main control board in the slot that originally housed a WMPT of the original root BBU.
- e. Remove the UBBPd from the original leaf BBU.
- f. Remove the original leaf BBU and replace it with a new BBU59X0. This subrack is used to deploy LTE and NR and function as the new root BBU.
- g. In the new root BBU, install the UMPTg, UBBPg, and the UBBPd that was removed from the original leaf BBU.
- h. Use a BBU interconnection signal cable to connect the new root and leaf BBUs. Connect the cable end with a "Master" color-coding label to the CI port on the UMPTg in the new root BBU, and the other end to that on the UMPTb in the new leaf BBU.

 NOTE

- The UMPTb is changed from LTE to UMTS.
- Root BBU deployment requires L*N co-MPT reconstruction.
- Modifying configuration data
The data reconfiguration procedure using MML commands is as follows:
 - a. On the GSM side, run the **RMV BTSCTRLNK** command to remove the inter-BBU interconnection link.
 - b. On the GSM side, run the **RMV BTSBRD** command to remove the UCIU from the original root BBU.
 - c. On the LTE side, run the **RMV CTRLNK** command to remove the inter-BBU interconnection link.
 - d. On the UMTS side, run the **RMV CTRLNK** command to remove the inter-BBU interconnection link.
 - e. On the LN co-MPT side, run the **ADD SUBRACK** command to add a root BBU (BBU59X0).
 - f. On the LN co-MPT side, run the **ADD BRD** command to add a UMPTg and a UBBPg for the root BBU.
 - g. On the LN co-MPT side, run the **ADD CTRLNK** command to add an inter-BBU interconnection link.
 - h. On the UMTS side, run the **ADD CTRLNK** command to add an inter-BBU interconnection link.
 - i. On the GSM side, run the **ADD BTSCTRLNK** command to add an inter-BBU interconnection link.

5 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

6 Reference Documents

This document uses the following references:

1. *Common Transmission Feature Parameter Description*
2. *Synchronization Feature Parameter Description*
3. *Common Clock Feature Parameter Description*
4. *RF Unit and Topology Management Feature Parameter Description*
5. *CPRI MUX Feature Parameter Description*
6. *Separate-MPT Base Station Initial Configuration Guide (MAE-Deployment-based)*
7. *Co-MPT Base Station Initial Configuration Guide (MAE-Deployment-based)*